sPHENIX Internal Note

Eta Edge Migration Study: Fake Rate from Fiducial Boundary Effects

Isolated Photon Cross-Section in p+p at $\sqrt{s} = 200$ GeV

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Abstract

This note documents a study of pseudorapidity edge migration effects in the sPHENIX isolated photon cross-section measurement (PPG12). Both the truth and reconstructed η fiducial cuts are set at $|\eta| < 0.7$, which allows truth photons produced outside the fiducial volume to have reconstructed clusters migrate inside, and vice versa. In the unfolding framework, inward-migrated clusters matched to truth photons outside $|\eta| < 0.7$ should be treated as fakes but currently enter the response matrix as signal in `RecoEffCalculator_TTreeReader.C`. We quantify this contamination for both single-interaction and double-interaction MC, finding a signal region (ABCD region A) contamination of $\sim$1.2% for single interaction and $\sim$0.3% for double interaction, with significantly larger effects in the non-tight ABCD regions used for background estimation.

Contents

1 Introduction

The PPG12 analysis measures the isolated prompt photon production cross-section in p+p collisions at $\sqrt{s} = 200$ GeV using the sPHENIX EMCal. The fiducial pseudorapidity acceptance is $|\eta| < 0.7$, corresponding to the barrel EMCal coverage. This cut is applied identically to both truth-level photons and reconstructed clusters.

Because the η boundary is a sharp geometric cut rather than a smooth efficiency edge, photons near $|\eta| \approx 0.7$ can migrate across the boundary between truth and reconstruction:

- **Inward migration (fakes):** Truth photons with $|\eta^{\text{truth}}| > 0.7$ produce reconstructed clusters with $|\eta^{\text{reco}}| < 0.7$. These enter the signal sample but have no corresponding truth entry within the fiducial volume.
- **Outward migration (losses):** Truth photons with $|\eta^{\text{truth}}| < 0.7$ produce reconstructed clusters with $|\eta^{\text{reco}}| > 0.7$, escaping the reconstructed sample.

In the current implementation of `RecoEffCalculator_TTreeReader.C`, inward-migrated clusters enter the response matrix as signal — i.e., they are treated as correctly matched truth-reco pairs. This is incorrect: their truth p_{T} falls outside the fiducial region, biasing the unfolding. This study quantifies the magnitude of this effect.

2 Method

A dedicated macro `EtaMigrationStudy.C` processes the full photon MC samples: `photon5` (truth p_{T} range 0–14 GeV), `photon10` (14–30 GeV), and `photon20` (30+ GeV). Each sample is weighted by its production cross-section.

Truth photons are registered with *no* η filter — all direct/fragmentation photons satisfying $E_{\text{T}}^{\text{iso,truth}} < 4$ GeV (cone $R_{\text{cone}} = 0.3$) are included regardless of η. Each truth–reco matched pair is then classified into four categories based on the fiducial boundary $|\eta| < 0.7$:

1. **Good:** both truth and reco inside $|\eta| < 0.7$.
2. **Fake (inward):** truth outside, reco inside.
3. **Lost (outward):** truth inside, reco outside.
4. **Both outside:** neither truth nor reco in fiducial volume.

The standard PPG12 ABCD classification is applied to each pair:

- Region A (tight + isolated): signal region.
- Region B (tight + non-isolated): background template.
- Region C (non-tight + isolated): background control.
- Region D (non-tight + non-isolated): background normalization.

For the double-interaction study, a toy vertex shift is applied: a random second vertex is drawn from the data vertex distribution, and the new vertex is computed as $z_{\text{vtx}}^{\text{new}} = (z_{\text{vtx}}^{\text{orig}} + z_{\text{vtx}}^{\text{rand}})/2$. Cluster kinematics are recalculated using the CEMC cylinder geometry ($R = 93.5$ cm):

$$z/r_{\text{new}} = z/r_{\text{old}} + \frac{z_{\text{vtx}}^{\text{old}} - z_{\text{vtx}}^{\text{new}}}{93.5\ \text{cm}}, \quad \eta_{\text{new}} = \sinh^{-1}(z/r_{\text{new}}), \quad E_{\text{T}}^{\text{new}} = E/\cosh(\eta_{\text{new}}). \tag{1}$$

3 Results

3.1 Overall Migration Rates

Table 1 summarizes the unweighted migration rates per MC sample. The fake (inward) rate is defined as the number of inward-migrated pairs divided by the total number of matched pairs, and similarly for the loss (outward) rate.

Table 1: Overall migration rates (unweighted) for single-interaction MC. Fake rate = inward / total; loss rate = outward / total.

Sample	Total	Good	Fakes	Losses	Both out	Fake rate	Loss rate
photon5	2361432	1523346	18490	13687	805909	1.20%	0.89%
photon10	532878	374380	3942	5822	148734	1.04%	1.53%
photon20	128595	107741	721	2193	17940	0.66%	1.99%

The fake rate decreases with increasing truth p_{T} range (from 1.20% to 0.66%), while the loss rate increases (from 0.89% to 1.99%). The asymmetry arises because higher-p_{T} photons have narrower shower profiles and more precise η reconstruction, but the steeply falling spectrum preferentially populates the inward direction at lower p_{T}.

3.2 Signal Region (A) Contamination

The physically relevant metric is the contamination of the ABCD signal region (region A: tight BDT + isolated). This is defined as:

$$\text{Contamination} = \frac{N_{\text{fakes}}^{\text{tight+iso}}}{N_{\text{good}}^{\text{tight+iso}} + N_{\text{fakes}}^{\text{tight+iso}}}. \tag{2}$$

This quantity directly measures the fraction of response matrix entries that are edge-migration fakes.

3.2.1 Single Interaction

Table 2 shows the signal region contamination per p_{T} bin for single-interaction MC.

The contamination is approximately flat at ∼1.2–1.3% across most of the p_{T} range but rises to 2.57% in the highest bin (32–36 GeV).

3.2.2 Double Interaction

Table 3 shows the equivalent for double-interaction (toy vertex shift).

The double-interaction contamination in region A is significantly *lower* than single interaction (0.34% vs. 1.24%). This counterintuitive result is explained in Section 3.4.

3.3 ABCD Region Breakdown

The edge migration does not affect all ABCD regions equally. The contamination varies strongly by ABCD region, particularly in the highest p_{T} bins where the effect is most pronounced.

Region C (non-tight + isolated) exhibits dramatically higher contamination, reaching up to 52% at 32–36 GeV for single interaction. This occurs because edge-migrated clusters have distorted

Table 2: Signal region (tight + isolated) contamination from η edge migration, single-interaction MC. Counts are cross-section weighted.

p_{T} bin (GeV)	Good (tight+iso)	Fakes (tight+iso)	Contamination
8–10	1.04e8	1.30e6	1.24%
10–12	3.71e7	4.43e5	1.18%
12–14	1.45e7	2.06e5	1.40%
14–16	6.28e6	7.87e4	1.24%
16–18	2.86e6	3.76e4	1.30%
18–20	1.35e6	1.79e4	1.31%
20–22	6.62e5	8.68e3	1.29%
22–24	3.41e5	4.42e3	1.28%
24–26	1.79e5	2.34e3	1.30%
26–28	9.19e4	1.61e3	1.72%
28–32	7.70e4	1.49e3	1.90%
32–36	2.11e4	5.57e2	2.57%
Overall	1.67e8	2.10e6	**1.24%**

Table 3: Signal region (tight + isolated) contamination from η edge migration, double-interaction MC (toy vertex shift).

p_{T} bin (GeV)	Good (tight+iso)	Fakes (tight+iso)	Contamination
8–10	7.81e6	2.58e4	0.33%
10–12	2.78e6	1.12e4	0.40%
12–14	1.00e6	3.37e3	0.33%
14–16	4.09e5	1.33e3	0.33%
16–18	1.79e5	3.19e2	0.18%
18–20	8.37e4	2.66e2	0.32%
20–22	4.00e4	2.13e2	0.53%
22–24	2.07e4	1.07e2	0.51%
24–26	1.10e4	0	0.00%
26–28	5.82e3	53	0.91%
28–32	4.87e3	1	0.02%
32–36	1.39e3	1	0.07%
Overall	1.24e7	4.27e4	**0.34%**

shower shapes: the η mismatch between truth and reco positions distorts the transverse shower profile, causing these clusters to preferentially fail the tight BDT selection and land in the non-tight category.

The signal region A contamination remains modest (1.2–2.6%) precisely because the BDT effectively rejects edge fakes — they carry anomalous shower shapes that the BDT was trained to discriminate against.

3.4 Double Interaction Enhancement

The overall *raw* inward fake rate (before ABCD classification) is strongly amplified by double interaction: 1.2% (single) vs. 13.7% (double), approximately an 11× amplification. The vertex displacement in double-interaction events shifts the reconstructed η of all clusters, dramatically increasing the probability of boundary crossing.

However, the signal region contamination moves in the *opposite* direction: 1.24% (single) → 0.34% (double). The explanation is that vertex-shifted clusters acquire additional shower shape distortion on top of the edge migration effect. The compounded distortion causes the tight BDT to reject these clusters even more aggressively, suppressing their entry into region A.

Figure 9 quantifies this by showing the fraction of ABCD-classified fakes (i.e., those passing common quality cuts) that leak into regions A and C, as a function of E_{T}. For single interaction, ∼50% of ABCD-classified fakes pass tight+iso (region A) and ∼40% land in region C. For double interaction, a *higher* fraction (∼70–85%) passes tight+iso — a seemingly counterintuitive result. The explanation is a *selection effect*: the common quality cuts reject 98.4% of double-interaction inward fakes (vs. 81% for single), acting as a much stronger pre-filter. The 1.6% of double fakes that survive are the "best" fakes — truth photons barely outside the boundary with small vertex shifts — and these naturally also tend to pass the tight BDT. The common cuts are doing the heavy lifting; their survivors are biased toward photon-like signatures. The absolute signal region contamination (Fig. 7 vs. Fig. 8) remains much lower for double interaction (0.34% vs. 1.24%) because the total number of ABCD-classified fakes is ∼60× smaller.

It should be noted that the double-interaction statistics are marginal: only 77 unweighted tight+iso fake entries contribute to the entire double-interaction signal region, limiting the precision of the contamination estimate.

3.5 Response Matrix Contamination

The response matrix in `RecoEffCalculator_TTreeReader.C` is filled for all tight+iso matched pairs. The 1.2% signal region contamination from Table 2 directly translates to 1.2% of response matrix entries having incorrect truth p_{T} (from outside the fiducial η region). These entries map reco p_{T} values to truth p_{T} values that should not participate in the unfolding, introducing a bias in the unfolded cross-section.

The effect is p_{T}-dependent: rising from ∼1.2% at low p_{T} to ∼2.6% at 32–36 GeV. This p_{T} dependence means the bias is not a simple overall normalization shift but distorts the shape of the unfolded spectrum.

3.6 ABCD Leakage Fractions: Single vs. Double Interaction

The ABCD signal leakage fractions c_B, c_C, c_D (defined as the ratio of truth-matched signal in regions B, C, D to that in region A) directly enter the background subtraction and purity correction. Figure 12 shows these fractions computed from the "good" signal (truth inside fiducial $|\eta| < 0.7$) for both single and double interaction.

- c_B **(tight, non-isolated):** Increases from ∼6–11% (single) to ∼11–29% (double), a factor ∼2–3×. The vertex shift in double-interaction events smears the topo-cluster isolation sum, causing isolated photons to fail the isolation cut and migrate from region A to B.

- c_C **(non-tight, isolated):** Comparable between single and double (∼2–8%), with no systematic amplification. The shower shape BDT is less sensitive to vertex shifts at the level relevant here.

- c_D **(non-tight, non-isolated):** Small in both cases ($< 2\%$), with modest increase for double interaction.

The dominant effect is the c_B amplification. Since the ABCD background estimate is $N^A_{\text{bkg}} = R \cdot B \cdot C/D$ with signal leakage corrections proportional to $(1 - c_B)(1 - c_C)/(1 - c_D)$, the 2–3× increase in c_B for double interaction significantly affects the corrected purity, particularly at high E_{T} where the leakage is largest.

4 Discussion

The ∼1.2% signal region contamination is a systematic effect, not a statistical fluctuation. At the highest p_{T} bin (32–36 GeV), the contamination reaches 2.6%, which is comparable in magnitude to other systematic uncertainties in the analysis (e.g., energy scale, BDT threshold variations).

Three approaches could address this:

1. **Asymmetric η cuts:** Apply a tighter reco η cut than truth (e.g., $|\eta^{\text{reco}}| < 0.6$ with $|\eta^{\text{truth}}| < 0.7$). This eliminates inward migration but sacrifices acceptance.

2. **Truth η check in response matrix:** Explicitly verify that the truth η of matched pairs satisfies $|\eta^{\text{truth}}| < 0.7$ before filling the response matrix. Outside-fiducial matches are counted as fakes rather than signal. This is the minimal fix.

3. **Systematic uncertainty:** Include the contamination as a systematic uncertainty on the unfolded cross-section, propagated through the full pipeline.

The region C contamination (up to 52% at high p_{T}) is a separate concern. Region C enters the ABCD background estimation through the R-factor: $R = N_C/N_D$. If a large fraction of region C entries are edge-migration fakes rather than genuine non-tight background, the R-factor and consequently the background subtraction could be biased.

For double interaction, the tight+iso selection provides effective rejection of edge-migrated clusters (contamination drops from 13.7% raw to 0.34% in region A). However, the ∼14% raw migration rate is large enough to significantly affect the non-tight regions used in background estimation.

5 Recommendations

1. **Short term:** Add the truth η check to the response matrix filling in `RecoEffCalculator_TTreeReader.C`. Matched pairs with $|\eta^{\text{truth}}| > 0.7$ should be counted as fakes, not signal.

2. **Medium term:** Evaluate the impact on the unfolded cross-section by running the full pipeline with and without the fix. The difference quantifies the bias introduced by the current implementation.

3. **Background estimation:** Evaluate and include the region C contamination as a systematic uncertainty on the ABCD R-factor.

6 Appendix: Plots

Figures 1–11 show the distributions produced by `plot_eta_migration.C`.

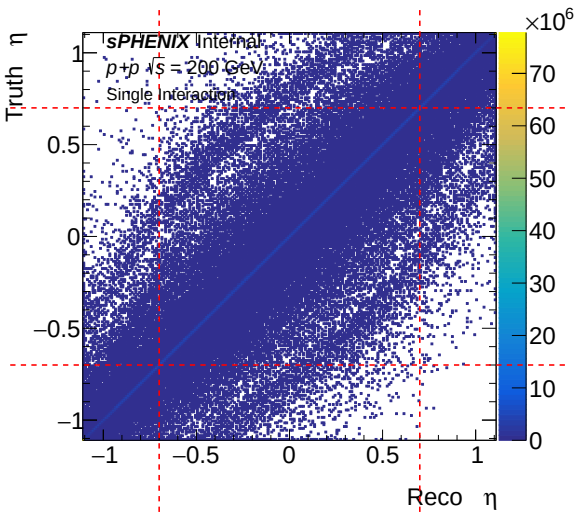

Figure 1: Migration matrix: truth vs. reco η classification for single-interaction MC.

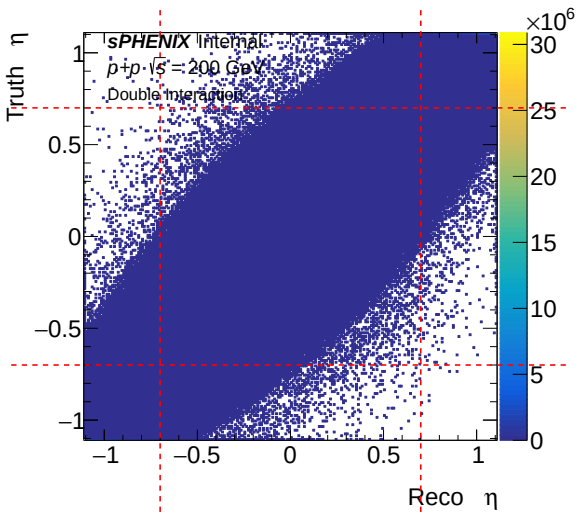

Figure 2: Migration matrix: truth vs. reco η classification for double-interaction MC (toy vertex shift).

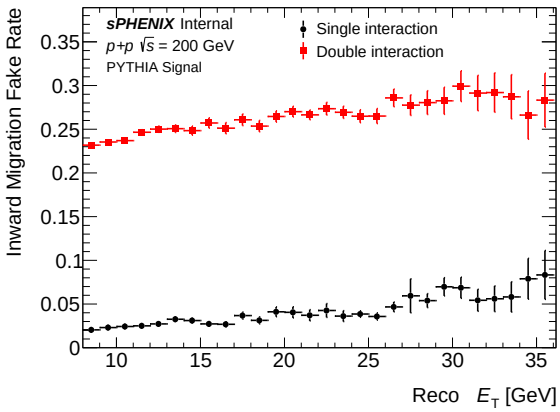

Figure 3: Inward fake rate vs. E_{T} for single and double interaction.

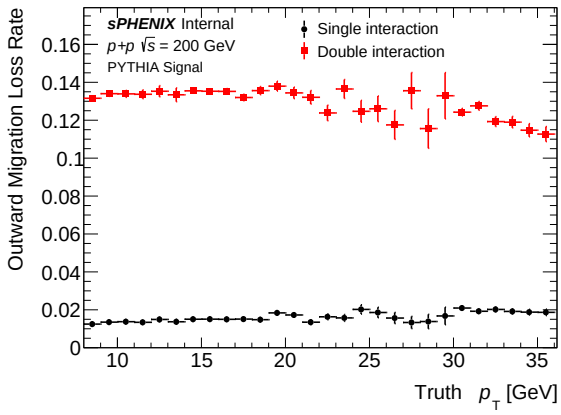

Figure 4: Outward loss rate vs. E_{T} for single and double interaction.

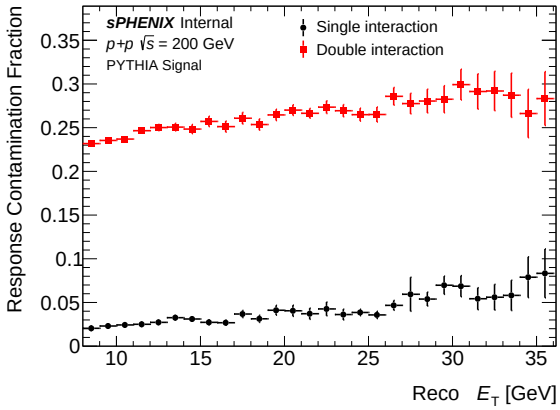

Figure 5: All-cluster inward migration fraction vs. E_{T} for single (∼2–8%) and double (∼25%) interaction. This is the raw rate before any ABCD selection.

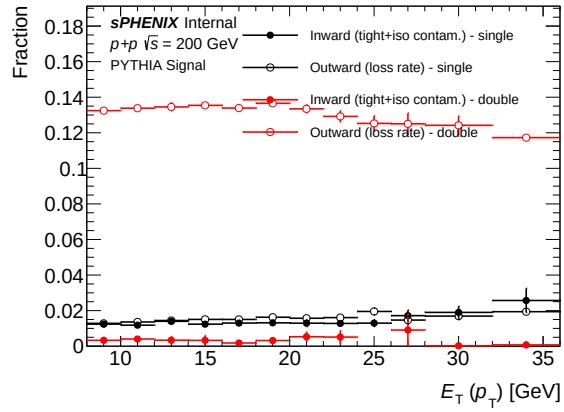

Figure 6: Response matrix impact: tight+iso inward contamination (filled circles) and outward loss rate (open squares) vs. E_{T}. Single interaction: ∼1.2% contamination and ∼1.5% loss. Double: ∼0.3% contamination but ∼13% loss.

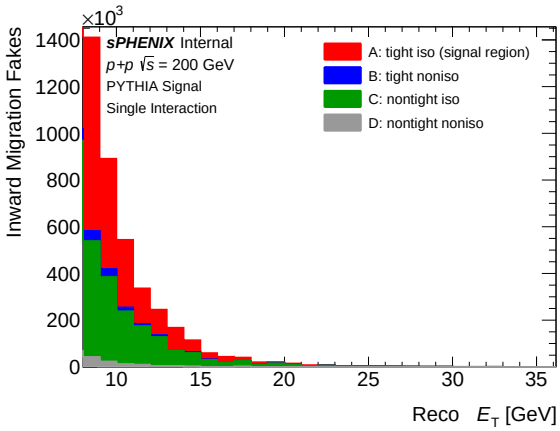

Figure 7: ABCD distribution of inward-migration fakes for single interaction. Region A (tight+iso) and C (nontight+iso) dominate.

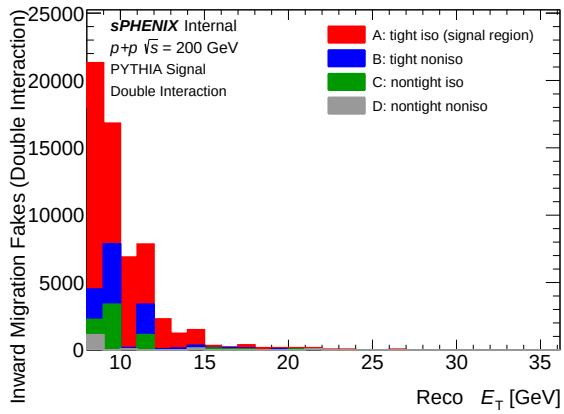

Figure 8: ABCD distribution of inward-migration fakes for double interaction. The total yield is ∼60× smaller than single interaction (Fig. 7), because 98% of double-interaction fakes fail common quality cuts.

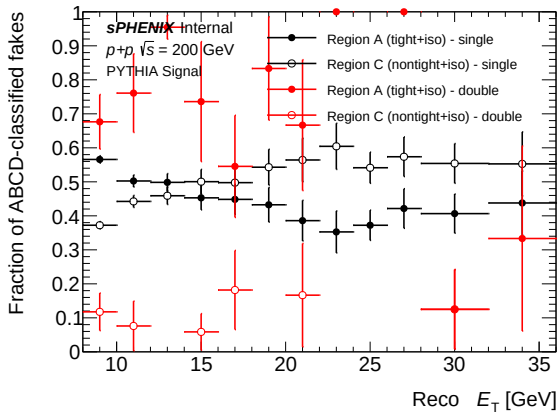

Figure 9: Fraction of inward-migrating fakes that leak into regions A (tight+iso, filled circles) and C (non-tight+iso, open squares) as a function of E_{T}, comparing single (black) and double (red) interaction. Single-interaction fakes have a higher probability of passing tight+iso, while double-interaction fakes are more efficiently rejected into region C.

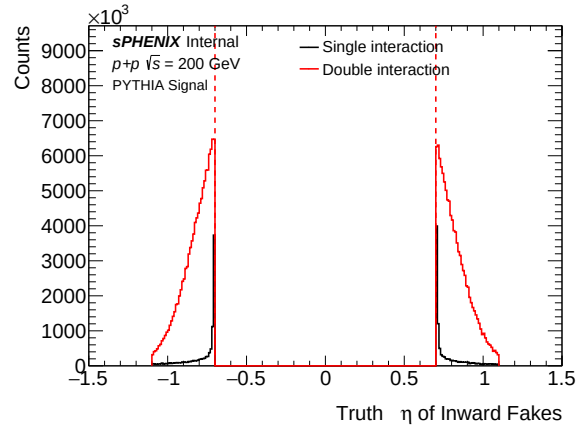

Figure 10: Truth η distribution of inward-migrated fakes.

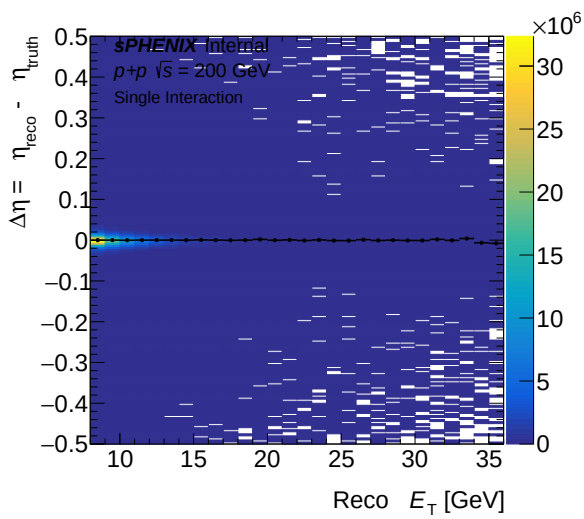

Figure 11: $\Delta\eta$ (truth $-$ reco) vs. E_{T} for edge-migrated clusters.

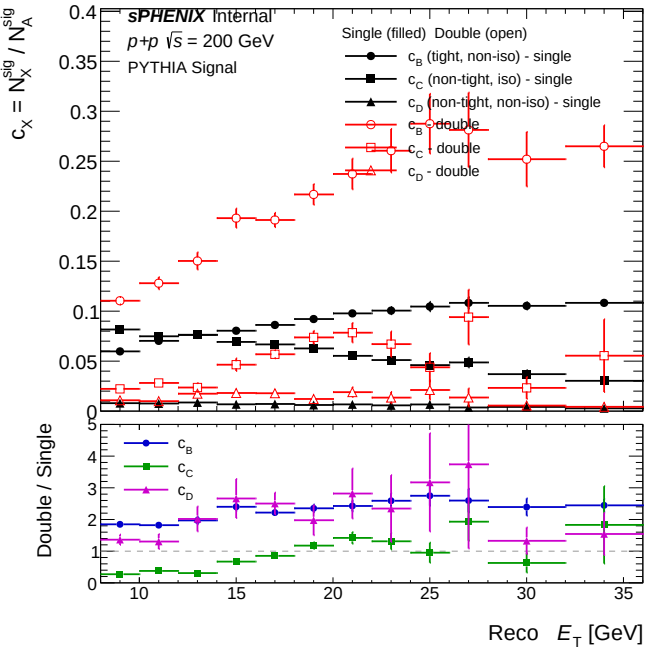

Figure 12: ABCD signal leakage fractions $c_X = N_X^{\mathrm{sig}}/N_A^{\mathrm{sig}}$ for single interaction (filled markers) and double interaction (open markers) as a function of E_{T}. Upper panel: absolute leakage fractions for regions B, C, and D. Lower panel: double-to-single ratio. The isolation leakage c_B increases by a factor of $\sim$2–3$\times$ under double interaction due to vertex-shift smearing of the isolation cone, while the shower-shape leakage c_C is comparable or slightly reduced.

7 Sideband Flow from Background-Only MC

The results so far quantify the impact of eta edge migration on the signal sample using photon MC. The PPG12 ABCD background estimate relies on three sideband regions (B, C, D) that are populated overwhelmingly by jet-induced fake clusters in data. To assess how fiducial boundary migration distorts these sideband populations — and therefore the background estimate itself — we run `EtaMigrationStudy.C` on the background-only MC with *every* truth particle registered as a candidate match (not just direct/fragmentation photons), using `cluster_truthtrkID` to find the dominant truth contributor per cluster — typically a decay photon from a π^0 or η in jet MC.

Two interaction scenarios are studied:

Single-interaction background jet20 ($21 < p_{\mathrm{T}}^{\mathrm{jet}} < 32$) and jet30 ($32 < p_{\mathrm{T}}^{\mathrm{jet}} < 42$) samples, cross-section weighted and merged. This matches the nominal PPG12 background MC.

Double-interaction background the full-GEANT pileup sample `jet12_double` ($p_{\mathrm{T}}^{\mathrm{jet}} > 10$), simulated with two overlapping p+p collisions at nominal luminosity. This represents the asymptotic "100% pileup" rate; in real data at 1.5 mrad crossing only ∼7.9% of clusters come from double interactions.

7.1 The NPB-Sentinel Issue

The PPG12 ABCD common preselection requires an NPB (non-photon-background) TMVA score > 0.5. The NPB model was trained only on clusters with $|\eta| < 0.7$; for clusters reconstructed outside the fiducial, the stored NPB score is the sentinel value -1. Consequently the NPB cut rejects *every* cluster outside the fiducial volume, which eliminates outward-migration candidates from the shower-shape sidebands before any flow measurement can be made. To restore the symmetry, the eta-migration analysis fills a parallel "no-NPB" histogram set in which the NPB common cut is skipped. The inflow (measured) side is essentially unchanged because NPB rejects almost no in-fiducial background clusters (the score is typically > 0.9 for real showers). The outflow side now carries meaningful statistics.

All plots in this section use the no-NPB classification.

7.2 Definitions

For each cluster with a valid truth match, we classify into one of three flow categories per ABCD region $X \in \{A, B, C, D\}$:

Native truth and reco both inside $|\eta| < 0.7$, cluster passes X. These are the clusters that *should* count in region X.

Inflow truth $|\eta| > 0.7$, reco $|\eta| < 0.7$, cluster passes X. Spurious additions to measured X — migration *into* the fiducial sideband.

Outflow truth $|\eta| < 0.7$, reco $|\eta| > 0.7$, cluster passes X (by shower-shape/isolation criteria, independent of reco η). Clusters that would have populated region X had the reco η been correctly inside fiducial — migration *out of* the fiducial sideband. Note that the analysis does *not* count these in the measured sideband, because the fiducial cut is applied on reco η.

We then define two flow rates:

$$r_{\text{in}}^{X}(E_{\text{T}}) = \frac{N_X^{\text{inflow}}}{N_X^{\text{native}} + N_X^{\text{inflow}}}, \tag{3}$$

$$r_{\text{out}}^{X}(E_{\text{T}}) = \frac{N_X^{\text{outflow}}}{N_X^{\text{native}} + N_X^{\text{outflow}}}. \tag{4}$$

r_{in}^{X} is the fraction of the *measured* region-X count that comes from migrated clusters. r_{out}^{X} is the fraction of clusters that *should* have populated region X but were lost to outward migration.

7.3 Single-Interaction Results

Table **??** shows the integrated flow rates for jet20+jet30 (single interaction). All rates are below 3%, with outflow and inflow of comparable magnitude. The tight BDT regions (A, B) show outflow slightly larger than inflow because the tight shower-shape cuts preferentially select clean in-fiducial clusters that happen to migrate outward; the edge-crossed clusters that migrate inward carry worse shower shapes and are rejected. Region C (non-tight, iso) is the region most affected by inward migration.

Table 4: Integrated inflow and outflow rates (jet20+jet30, single interaction) in the no-NPB classification.

Region	$\langle r_{\text{in}} \rangle$	$\langle r_{\text{out}} \rangle$
A: tight + iso	1.01%	1.80%
B: tight + non-iso	0.91%	1.95%
C: non-tight + iso	2.70%	1.65%
D: non-tight + non-iso	1.22%	1.33%

7.4 Double-Interaction Results

The full-GEANT double-interaction sample `jet12_double` shows *dramatically* larger migration rates: the vertex shift from the overlap of two collisions pushes cluster η by $\Delta\eta \sim -\Delta z/(R_{\text{EMCal}} \cosh \eta) \sim 0.2$ on top of the single-interaction cluster-position resolution. Integrated rates reach several tens of percent in the tight-excluding non-tight regions (C, D), and the inflow dominates strongly over the outflow: the measured sideband populations are contaminated $\sim$ 30–40% by pileup-migrated clusters at high E_{T}. This is far larger than the single-interaction effect and drives the bulk of the pileup-related R-factor bias discussed below.

Note that `jet12_double` represents the pure double-interaction limit. At the nominal 1.5 mrad crossing angle, the cluster-weighted pileup fraction is $\sim$7.9% (from `calc_pileup_range.C`), so the effective bias in data is the weighted average $r^{\text{eff}} \approx 0.921\, r^{\text{single}} + 0.079\, r^{\text{double}}$.

7.5 Impact on the ABCD R-Factor

The ABCD background estimate uses $R = N_B \cdot N_C/N_D$ to project sideband counts into the signal region (with small signal-leakage corrections applied at the next level). Because inflow is present in all three non-signal regions, a substantial portion of the single-interaction bias cancels in the ratio. Figure **??** shows $R^{\text{meas}}/R^{\text{native}}$ where R^{meas} uses the measured sideband populations $N_X^{\text{meas}} = N_X^{\text{native}} + N_X^{\text{inflow}}$ and R^{native} uses N_X^{native} only.

For single-interaction jet MC, the R-factor bias is 4% across most of the analysis range, rising to $\sim$ 10% in the top two p_{T} bins where statistics are marginal. For pure double-interaction (`jet12_double`), the bias grows rapidly with E_{T} and reaches $\times 2$–$\times 2.5$ at $E_{\mathrm{T}} \sim$ 20–25 GeV. In the blended data scenario with 7.9% pileup, the effective bias is $1.0 + 0.079 \times (R^{\mathrm{d}}/R^{\mathrm{n}} - 1)$, which translates to roughly $\sim$ 6–12% at mid-E_{T} and potentially larger at high-E_{T}. This is at the same order as PPG12's existing sideband systematic uncertainties and warrants a follow-up systematic evaluation.

7.6 Figures

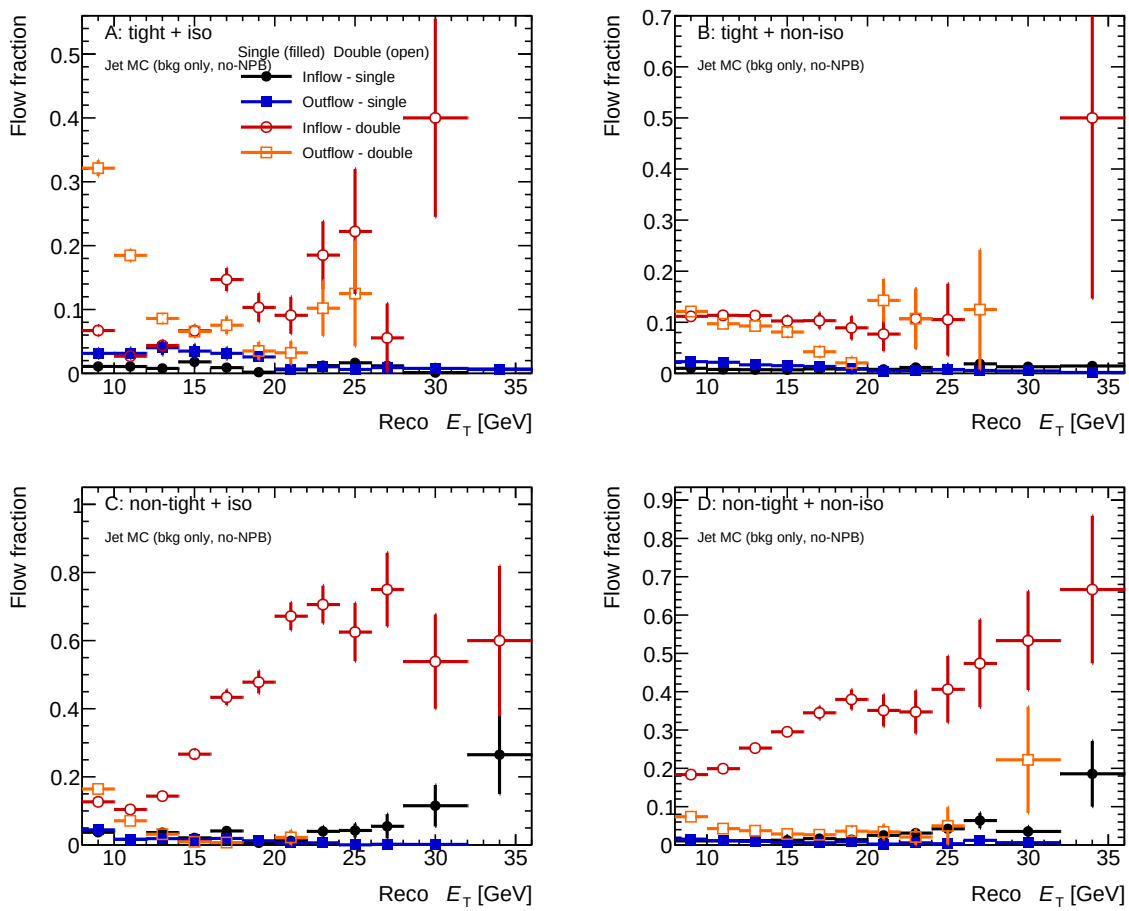

Figure 13: Inflow and outflow rates (no-NPB classification) vs. reco E_{T} per ABCD region in jet MC. Single interaction (filled black / blue markers) produces rates 3%; the full-GEANT `jet12_double` sample (open red / orange markers) drives much larger asymmetric inflow, reaching $\sim$ 30–70% in regions C and D at high E_{T}. The asymmetry $r_{\mathrm{in}} > r_{\mathrm{out}}$ in the double-interaction case reflects the vertex-shift bias: pileup reconstructs clusters at a different η than the leading truth particle.

7.7 Summary

- For single-interaction background MC the eta edge produces comparable inflow and outflow in every ABCD sideband (3%), and the residual R-factor bias is 4% across the analysis range.

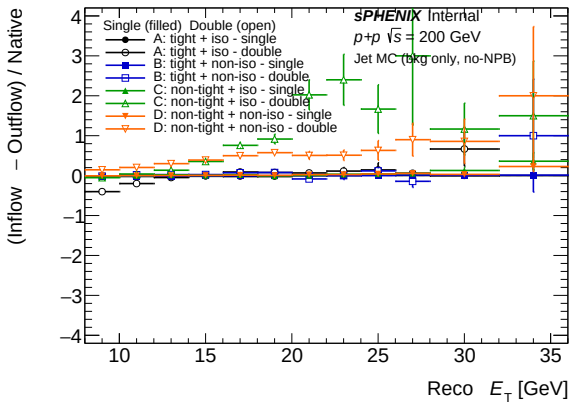

Figure 14: Net sideband bias $(N^{\text{inflow}} - N^{\text{outflow}})/N^{\text{native}}$ vs. reco E_{T} per region. Single interaction is consistent with zero at 2%; double interaction drives significant positive bias in regions C and D at high E_{T}.

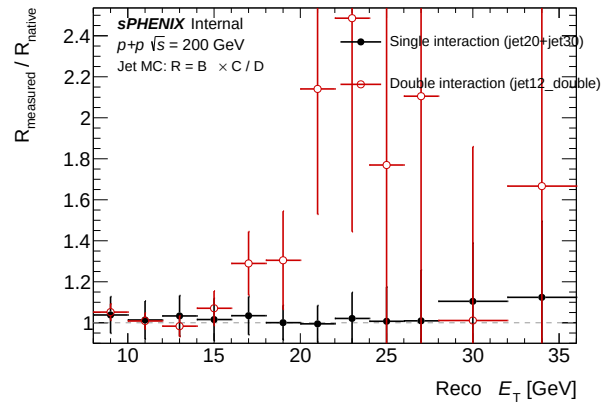

Figure 15: Ratio $R^{\text{meas}}/R^{\text{native}}$ of the ABCD R-factor using measured $(N^{\text{native}} + N^{\text{inflow}})$ vs. native populations. Single interaction shows $< 10\%$ bias; pure double interaction (`jet12_double`) shows a bias up to $\times 2.5$ at mid-E_{T}.

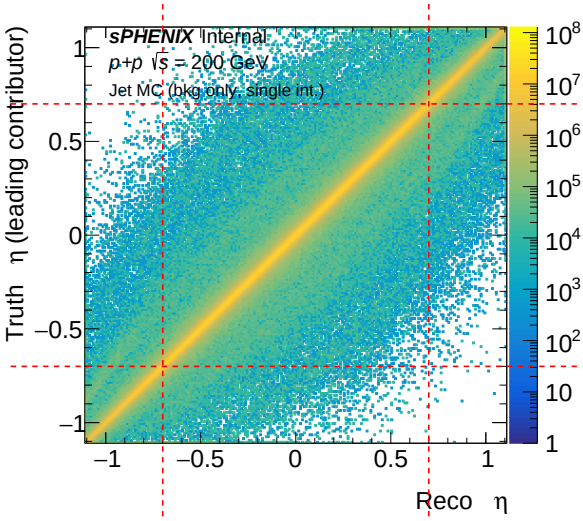

Figure 16: Migration matrix (reco η vs. truth η of the dominant contributor) for single-interaction jet MC. Narrow diagonal with small off-diagonal tails across the $|\eta| = 0.7$ boundary (red dashed). Color axis is logarithmic.

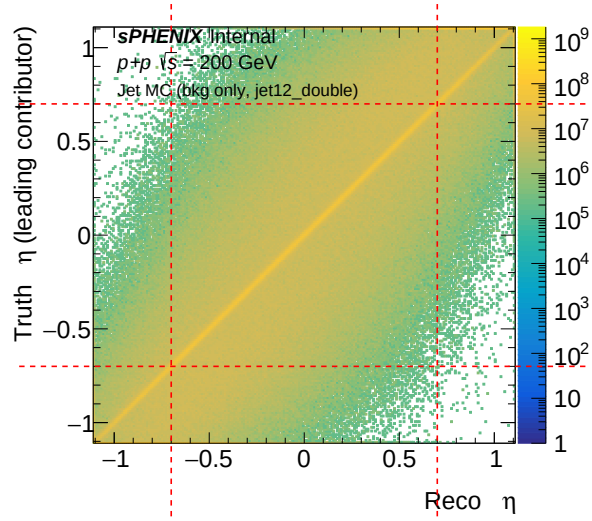

Figure 17: Migration matrix for the full-GEANT double-interaction sample `jet12_double`. The diagonal is visibly broader and extensive off-diagonal migration appears near and across the $|\eta| = 0.7$ boundary.

- For pure double-interaction MC (`jet12_double`) the inflow dominates strongly (tens of percent), and the R-factor bias reaches $\times 2$–2.5 at mid E_{T}.
- Applied with the measured 7.9% pileup fraction at 1.5 mrad, the effective R-factor bias in data is at the $\sim$ 5–15% level, comparable to other PPG12 ABCD systematic uncertainties and motivating a dedicated systematic evaluation.